



SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)

Syllabus for

B.S. (Hons) Computer Science

B.S. (Hons with Research) Computer Science

(Specialization in Artificial Intelligence)

(Applicable w.e.f the Academic year 2024-2025)

Department of Mathematics and Computer Science

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SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

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DEPARTMENT OF MATHEMATICS & COMPUTER SCIENCE

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Research) Computer Science
(Specialization in Artificial Intelligence)

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INTRODUCTION

Department of Mathematics and Computer Science is offering an Undergraduate program in the Major area of Computer Science viz: Three Year B.Sc. Computer Science, Four-year B.S Computer Science (Hons) with specialization in Artificial Intelligence (AI) and Four-year B.S Computer Science (Hons) specialization in Artificial Intelligence (AI) with Research.

The department also offers Minor courses in the following areas :

- Mathematics
- Data Science
- Actuarial Science

Students are admitted to this program after completing the 12th class under CBSE (or equivalent) boards with 10+2 in Mathematics from any state of India. In the first four semesters, essential aspects of Basic understanding of Programming are provided through courses such as Problem solving with Computer, Data Structures and Algorithms, Software Lab in C - Part-I and Part II, Operating Systems, Database Management System, Object Oriented Programming Concepts, Computer Organization System and Software Lab in C++. These courses give a great foundation for students in the area of Computer Science. In the last four semesters, they specialize through an AI stream of courses for an Honors degree. At the end of the Four Year program, students have a better understanding of both Computer Science and Artificial Intelligence.

Students after completing the program either join Research or continue for M.Sc. Computer Science or AI and Computing or some professional courses like M.Tech. or take up jobs in various fields including software development and AI research. The B.S in Computer Science Honors syllabus aims to achieve the following objectives.

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Program Educational Objectives (PEOs)

PEO1: Critical Thinking

Be developing a diverse range of critical thinking skills, including problem-solving, logical reasoning, data analysis, and abstraction.

PEO2: Effective Communication

Be able to convey complex concepts clearly, collaborate with diverse teams, actively listen to client needs, and adapt communication to various stakeholders, ensuring effective solutions and successful project outcomes.

PEO3: Social Interaction

acquire a comprehensive set of social interaction skills, including active listening, effective communication, collaboration, problem solving, time management, networking, conflict resolution, leadership, adaptability, professionalism, cultural sensitivity, self-awareness, presentation skills, and mentorship.

PEO4: Effective Citizenship

gain knowledge, critical thinking skills, and a sense of social responsibility needed to become effective citizens who can contribute positively to their communities and engage in informed civic discourse.

PEO5: Ethics

develop the skills and knowledge they need to be ethical citizens and professionals, to become critical thinkers, and to develop their own moral compass.

PEO6: Environment and Sustainability

Develop a profound understanding of the intricate human-environment relationship, acquire skills to tackle environmental challenges, and become engaged advocates for sustainability and environmental protection.

PEO7: Self-directed and life-long learning

Develop self-directed and lifelong learning skills through the opportunities to make choices, effective learning strategy instruction, and the establishment of a supportive learning environment.

Program Specific Outcomes:

By end of the course, students will have

S. No	Course	Outcome
1	Language courses	The language courses are introduced to train the students in the skill set of writing and speaking coherently and convincingly about a given topic. These skills are absolutely essential these days in the work environment.
2	Awareness	Awareness Courses are novel and unique to this programme. These courses help the student to relate to their cultural roots on one hand and understand the present scenario in the society with all its complexities and composition. It builds the motivation in the student to imbibe and live moral values in the society.
3	Computer Organization and Design	Courses like “Computer Organization and Design”, “Introduction to Parallel Computing” provide a complete understanding of computer organization, its latest trends, and how to achieve performance on such systems.
4	Theoretical Computer Science	Courses like “Problem Solving in Computer”, “Data Structures and Algorithms”, and “Design and Analysis of Algorithms” lay down the foundations required and the skills needed to develop algorithms and to perform the analysis.
5	Web and Computer Networks	Courses like “Computer Networks”, “Web programming” provide the internals behind the working of computer networks and the World Wide Web.
6	Mathematics	Courses like Foundations in Algebra and Calculus, Probability and Statistics, Linear Algebra, and Discrete Mathematics lay the mathematical and statistical foundation for learning computer science.
7	Databases	Courses like “Database Management Systems” and “SQL Programming” provide knowledge of database theory and practicals.

8	Programming Skills	Students are also given sufficient training in programming languages like C, C++, Java, SQL, and Python.
9	Research Project / Project work	The project aims to expose the students to the real-world challenges. The student is expected to present the work to an internal committee in the form of project viva-voce. A separate grade for the project is shown on the grade card.
10	Statistics	The probability and statistical techniques help the students analyze, understand, and interpret the data. The statistics courses help the students understand small datasets as well as big data.
11	Data Science	Courses like “Statistics for Data Science”, “Linear Algebra for Data Science” and “Optimization for Machine Learning” give sufficient skills in understanding machine learning algorithms and to make the students future ready.
12	AI	It helps the student to improve his cognitive ability in understanding human thinking and apply it to the machines, making the machine behave naturally, and perform tasks that are otherwise difficult at a human level.

Scheme of Assessment:

The examination system has got 2 components:

- a. Continuous Internal Evaluation (CIE)
- b. Summative Test (ST)

CIE for each subject consists of quizzes, assignments or oral tests. These tests are conducted at regular intervals during the semester. These help the student keep abreast with the pace of the course and to reinforce learning. The variety of tests enables both the students and the teachers to evaluate the learning outcomes and performance in various forms of expression. Each component of CIE has its own role and importance in the overall evaluation.

ESE on each subject is conducted once at the end of the semester which is comprehensive covering the entire syllabus of the course.

SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING**(DEEMED TO BE UNIVERSITY)****SCHEME OF INSTRUCTION & EVALUATION****B.S (Honors) Computer Science / B.S (Honors with Research) Computer Science****(Applicable w.e.f the Academic year 2024-2025)**

Paper Code	Title of the Paper	Category	Credits	Hours (L-T-P)	Modes of Evaluation	Types of Papers	Maximum Marks
Semester I							
UELS-101	English Language Skills-I	AEC	3	3-0-0	IE	L	100
	2nd Language	AEC	3	3-0-0	IE	L	100
AWR-100	Awareness Course-I: Sai Education for Transformation (Based on Life and Teachings of Bhagawan Baba	AWR	2	2-0-0	I	L	50
UENT-001	Environmental Studies	MDC	3	3-0-0	I	L	100
UDSC-101	Calculus	IDM	4	3-1-0	IE	L+T	100
UCSH-101	Problem Solving with Computer	DSC	4	4-0-0	IE	L	100
UCSH-102	Software Lab in C - Part I	DSC	3	0-0-6	I	P	100
UCSH-103	Software Lab in Python - I	SEC	1	0-0-2	I	P	50
			23 Credits	27 Hours			700 Marks

Paper Code	Title of the Paper	Category	Credits	Hours (L-T-P)	Modes of Evaluation	Types of Papers	Maximum Marks
Semester II							
UELS-201	English Language Skills-II	AEC	3	3-0-0	IE	L	100
	2nd Language	AEC	3	3-0-0	IE	L	100
AWR-200	Awareness Course-II: Unity of Religions	AWR	2	2-0-0	I	L	50
	Indian Constitution	MDC	3	3-0-0	I	L	100
UDSC-201	Probability and Statistics	IDM	4	3-1-0	IE	L+T	100
UCSH-201	Data Structures and Algorithms	DSC	4	4-0-0	IE	L	100
UCSH-202	Software Lab in C - Part II	DSC	3	0-0-6	I	P	100
UCSH-203	Software Lab in Python-II	SEC	1	0-0-2	I	P	50
			23 Credits	27 Hours			700 Marks

Semester III							
UELS-301	English Language Skills - III	AEC	3	3-0-0	IE	L	100
AWR-300	Awareness Course-III: Study of Classics-I: Ramakatha Rasavahini	AWR	2	2-0-0	I	L	50
	Multidisciplinary Course	MDC	3	3-0-0	I	L	100
UDSC-301	Statistics for Data Science	IDM	4	3-1-0	IE	L+T	100
UCSH-301	Computer Organization and Design	DSC	4	4-0-0	IE	L	100

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UCSH-302	Discrete Mathematics for Computer Science	DSC	4	4-0-0	IE	L	100
UCSH-303	Software Lab in Data Visualization	SEC	2	0-0-4	I	P	50
			22 Credits	24 Hours			600 Marks

Paper Code	Title of the Paper	Category	Credits	Hours (L-T-P)	Modes of Evaluation	Types of Papers	Maximum Marks
Semester IV							
AWR-400	Awareness Course-IV: Study of Classics-II: Bhagawatha Vahini	AWR	2	2-0-0	I	L	50
UDSC-401	Linear Algebra for Data Science	IDM	4	3-0-2	IE	L+P	100
UCSH-401	Database Management Systems	DSC	4	4-0-0	IE	L	100
UCSH-402	Object Oriented Programming Concepts	DSC	4	4-0-0	IE	L	100
UCSH-403	Software Lab in C++	DSC	3	0-0-6	I	P	100
UCSH-404	Software Lab in SQL	SEC	2	0-0-4	I	P	50
			19 Credits	25 Hours			500 Marks

Semester V							
AWR-500	Awareness Course-V: Eternal Values for the Changing World	AWR	2	2-0-0	I	L	50
UDSC-501	Optimization for Machine Learning	IDM	4	3-0-2	IE	L+P	100

UDSC-502	Data Mining and Machine Learning	IDM	4	3-0-2	IE	L+P	100
UCSH-501	Operating Systems	DSC	4	4-0-0	IE	L	100
UCSH-502	Computer Networks	DSC	4	4-0-0	IE	L	100
UCSH-503	Software Lab in Java	DSC	3	0-0-6	I	P	100
UCSH-504	Cyber Security	SEC	3	3-0-0	I	L	100
			24 Credits	29 Hours			650 Marks
Paper Code	Title of the Paper	Category	Credits	Hours (L-T-P)	Modes of Evaluation	Types of Papers	Maximum Marks
Semester VI							
AWR-600	Awareness Course-VI: Life and its Quest	AWR	2	2-0-0	I	L	50
UDSC-601	Big Data Analytics	IDM	4	4-0-0	IE	L	100
UCSH-601	Elective-I	DSE	4	4-0-0 (3-0-2)	IE	L	100
UCSH-602	Software Engineering	DSC	4	4-0-0	IE	L	100
UCSH-603	Design and Analysis of Algorithms	DSC	4	4-0-0	IE	L	100
UCSH-604	Software Lab in Web Programming	DSC	4	0-0-8	I	P	100
UCSH-605	Mini-Project/ Internship	PW	2	0-0-4	I	P	50
			24 Credits	30 (31) Hours			600 Marks
	Grand Total		135 Credits	162 (163) Hours			3750 Marks

Note: Students who wish to exit after successful completion of the sixth semester shall be awarded **B.Sc. Computer Science** degree.

Students who have a CGPA of 7.5 or above up to sixth semester and wish to earn **B.S(Hons) Computer Science with Research** must undertake a Research Project/Dissertation of 12 credits which starts from the seventh semester. Students with a CGPA less than 7.5 will also take a Project of 12 credits in lieu of the Research Project/Dissertation and will be awarded a **B.S (Hons) Computer Science** degree.

Paper Code	Title of the Paper	Category	Credits	Hours (L-T-P)	Modes of Evaluation	Types of Papers	Maximum Marks
Semester VII**							
AWR-700	Awareness-VII	AWR	2	2-0-0	I	L	50
UDSC-701	Cloud Computing	IDM	4	4-0-0	IE	L	100
UCSH-701	Elective-II	DSE	4	4-0-0 (3-0-2)	IE	L (L+P)	100
UCSH-702	Elective-III	DSE	4	4-0-0 (3-0-2)	IE	L (L+P)	100
UCSH-703	Linux System Programming	DSC	4	4-0-0	IE	L	100
UCSH-704	Research Methodology	RW	2	2-0-0	I	L	50
UCSH-803	Research Project/ Project Work Interim Report	RW/PW	-	0-0-6	I	-	50
			20 Credits	26 (28) Hours			550 Marks

Note : ** All the Minor courses will be completed by the 7th Semester.

Semester VIII							
AWR-800	Awareness-VIII	AWR	2	2-0-0	I	L	50
UCSH-801	Elective-IV	DSE	4	4-0-0 (3-0-2)	IE	L (L+P)	100
UCSH-802	Formal Languages	DSC	4	4-0-0	IE	L	100
UCSH-803	Research Project/ Project Work	RW/PW	12	0-0-18	I	D	150
			22 Credits	28 (29) Hours			450 Marks
	GRAND TOTAL		177 credits	216/220 Hours (max)			4750 Marks

Continuous Internal Evaluation (CIE) & End Semester Examination (ESE)

Categories of Courses		Modes of Evaluation	
Indicator	Legend	Indicator	Legend
AEC	Ability Enhancement Course	IE	CIE and ESE ; ESE single evaluation
DSC	Discipline Specific Core	I	Continuous Internal Evaluation (CIE) only
DSE	Discipline Specific Elective	E	End Semester Examination (ESE) only
SEC	Skill Enhancement Course		
MDC	Multidisciplinary Course		
IDM	Inter Disciplinary Minor		
RW	Research Work		
PW	Project Work		
AWR	Awareness Course		
		Note: 'I' does not connote 'Internal Examiner' 'E' does not connote 'External Examiner'	

PS: Please refer to guidelines for 'Modes of Evaluation for various types of papers', and 'Viva-voce nomenclature & scope and constitution of the Viva-voce Boards'.

Notes:

- 1. Specialization:** A student can earn the specialization in the degree title by completing any four electives from the stipulated list of specialization electives.

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2. **Mini Project:** The Mini-Project will be undertaken during the sixth semester by the candidate. This could be based on an internship (taken online/on-campus) with an industry or a field work etc., with a mentoring faculty from the department.

Students will be asked to make a presentation along with a submission of the report of the work done towards the end of the second semester i.e., within one week before the last working day of the semester. This will be evaluated internally by a panel of a minimum two faculty of the department constituted by HoD/Associate HoD. The total marks for the mini-project would be for 50 marks which will be equally distributed between the presentation and the (online) report submitted.

3. **Research Project (UCSH-803) / Project Work (UCSH-803)**

Research Project (UCSH-803): A Student who earns a CGPA of 7.5 and above in the first 6 semesters may take up a research project. The research project is for 12 Credits and evaluated for 200 Marks. Evaluation of the project is Internal by the project committee appointed by the HoD. The Evaluation method is given below:

- Evaluation will be done in THREE parts.
- First Evaluation is through interim evaluation for 50 marks done towards the end of 7th semester.
- The second evaluation component is through a project work presentation by the student to the project committee nominated by the Head of the Department (HoD/A.HoD) towards the end of the 8th semester, followed by a Viva Voce. This will be evaluated for 100 marks.
- Third Evaluation is through the project report submitted by the student to the guide. The project supervisor/an internal member will evaluate the project report for 50 marks.
- Project Committee: A student doing a project must have a project supervisor allotted by the HoD. The project supervisor and another one or two members of the Department nominated by the HoD constitute the Project Committee.

Project Work (UPW-800): A Student who does not earn a CGPA of 7.5 and above in the first 6 semesters may take up project work. The project work is for 12 Credits and evaluated for 200 Marks. Evaluation of the project is the same as URW-800.

4. **Discipline Specific Electives (DSEs) and Specialization:** The Department offers several interesting electives which are essential for applying Computer Science skills in various domains of IT. Among these DSEs, any one who completes four AI category DSE courses will earn the specialization degree.

DSEs for Artificial Intelligence Specialization

1. **UCSH-AI:** Artificial Intelligence [4 Credits]
2. **UCSH-DL:** Deep learning [4 Credits]
3. **UCSH-MLO:** Machine Learning Operations [4 Credits]
4. **UCSH-NLP:** Natural Language Processing [4 Credits]

- 5. Skill Enhancement Courses (SECs):** The department offers a few SECs to develop hands on skills and awareness which can be handy or supplementary to trendy areas in CS such as Data Science or Cyber Security

List of SECs

1. Software Lab in Python-I
2. Software Lab in Python-II
3. Software Lab in Data Visualization
4. Software Lab in SQL
5. Cyber Security

Note:

- The Choice of Electives and Streams of Specialization offered in a semester are subject to the discretion of the Head of the Department
- Specified Syllabus & Schema of Instruction and Evaluation for 3rd and 4th years is subject to revision based on future BoS recommendations
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Scheme of Evaluation and Instructions for Skill Enhancement Courses

Paper Code	Title of the Paper	Category	Credits	Periods	Modes of Evaluation	Types of Papers	Maximum Marks
Semester I							
UCSH-103	Software Lab in Python-I	SEC	1	0-0-2	I	P	50
Semester II							
UCSH-203	Software Lab in Python-II	SEC	1	0-0-2	I	P	50
Semester III							
UCSH-303	Software Lab in Data Visualization	SEC	2	0-0-4	I	P	50
Semester IV							
UCSH-404	Software Lab in SQL	SEC	2	0-0-4	I	P	50
Semester V							
UCSH-504	Cyber Security	SEC	3	3-0-0	I	L	100
			9 credits	15 Periods			250 Marks

6. Multi-Disciplinary Courses (MDCs): A student is expected to undertake a MDC elective that is outside his major discipline. The Department of Mathematics and Computer Science (DMACS) offers the following MDCs for undergraduate students.

Scheme of Evaluation of Multidisciplinary Course

Paper Code	Title of the Paper	Course Category	Credits	Periods L-T-P	Mode of Evaluation	Max. Marks
	Fundamentals of Statistics with Excel	MDC	3	2-0-1	I	100
	The Language of Mathematics	MDC	3	3-0-0	I	100
	Fundamentals of Actuarial Mathematics	MDC	3	3-0-0	I	100

7. Minor Disciplines : Department of Mathematics and Computer Science (DMACS) offers the following Minor Disciplines for Undergraduate students with required eligibility to earn a Minor Degree:

1. Mathematics
2. Computer Science
3. Actuarial Science
4. Data Science

However the Minor degree or courses will be offered subject to available logistics such as faculty, timetable slots etc.

Minor Certification: To earn a minor certification in a specific discipline, a student has to earn a minimum of 12 credits in 3 years of B.S program or 16 credits in 4 years of B.S Hons program in that particular discipline. It is to be noted that, the minimum credits earned must be from courses which are of difficulty level 300 or above. A student majoring in a particular discipline cannot obtain a minor in the same discipline.

The courses from any of the Minor disciplines will be offered based on the discretion of the HoD of DMACS.

Semester-I

Course Title	Problem Solving with Computer	L	T	P	C	Periods
Course Code	UCSH-101	4	0	0	4	56
Course Category	DSC					
Course Objectives: This course is designed to introduce students to algorithmic thinking. It introduces the fundamental concepts in algorithm design and how to apply these concepts in order to develop algorithms to solve specific problems.						
Course Outcome: Develop the skill set to 1. Think in the algorithmic way to solve simple computational problems 2. Choose the appropriate control structures necessary to solve specific problems 3. Formulate simple algorithms for arithmetic and logical problems 4. Test and execute the programs and correct syntax and logical errors 5. Implement conditional branching, iteration and recursion 6. Decompose a problem into functions and synthesize a complete program using divide and conquer approach 7. Use arrays and a few structures to formulate algorithms						
Prerequisites for the Course: 10+2 School Level Mathematics						
Unit	Topic	Contents				Periods
1	Introduction to Problem Solving and Fundamental Algorithms	Introduction, Problem Solving aspect, Top-down design, Foundation in efficiency and analysis of algorithms, Concept of a Variable, Iterative computational examples				11
2	Factoring Methods and Array techniques	Square root of a number, Greatest Common Divisor, Prime numbers, Prime factors, Raising a number to a large power, Fibonacci numbers, Simple Array operations, Removal of Duplicates, Array partitioning				11
3	Merging, Sorting and Searching	Two-way merge, Sorting by selection, Sorting by exchange, Sorting by insertion, Sorting by diminishing increment, Binary Search				11
4	Text Processing and Pattern Searching	Length adjustment, Left and right justification, Keyword search				12

5	Recursive Algorithms	Recursive Quicksort, Towers of Hanoi, Sample generation, Combination generation, Permutation generation	11
		Total	56

Key Text:

1. R. G. Dromey, How to Solve it by Computer, Pearson (Indian Subcontinent Low price edition), 2007. Chapters: 1 to 6, 8

Suggested Reading:

2. Jean-Paul Tremblay And Richard B. Bunt, Introduction to Computer Science, An Algorithmic Approach, McGraw Hill, Second International Edition, 1981.

Course Title	Software Lab in C - Part I	L	T	P	C	Periods
Course Code	UCSH-102	0	0	6	3	84
Course Category	DSC					

Course Objectives: This lab course introduces the basics of the C language to students. It would enable a student to implement algorithms for simple problems as programs in C thereby improving algorithmic thinking.

Course Outcome: Develop the skill set to

1. Formulate the algorithms as C programs for simple problems
2. Translate given algorithms to a working and correct program
3. Ability to correct syntax errors as reported by the compiler
4. Ability to identify and correct logical errors encountered at run time
5. Ability to write iterative as well as recursive programs
6. Ability to represent data in arrays and simple structures

Prerequisites for the Course: 10+2 School Level Mathematics

Unit	Topic	Contents	Periods
1	If-else, for & while loops	Basic programs for learning syntax, Maximum of n numbers, Printing different patterns of '*' based triangles	12
2	Iterative Programs	Square root of a number, Greatest Common Divisor, Prime numbers, Prime factors, Raising a number to a large power, Fibonacci numbers, Simple Array	18

		operations, Removal of Duplicates, Array partitioning.	
3	Merging, Sorting and Searching	Two-way merge, Sorting by selection, Sorting by exchange, Sorting by insertion, Sorting by diminishing increment, Sorting by partitioning, Binary Search.	18
4	Recursive Programs	Recursive Quicksort, Towers of Hanoi, Sample generation, Combination generation, Permutation generation	18
5	String Manipulation programs	Length of a string, String comparison, Palindrome check, Case changing, Length adjustment, Left and right justification, Keyword search.	18
		Total	84
References: 1. Richard F. Gilberg and Behrouz A. Forouzan, Computer Science - A Structured Programming Approach using C, ISBN-13: 978-0534491321, ISBN-10: 9780534491321, 2005. 2. Dennis Ritchie, The C Programming Language, 2nd edition, Prentice Hall India Learning Private Limited, 1990			

Semester - II

Course Title	Data Structures and Algorithms	L	T	P	Credits	Periods
Course Code	UCSH-201	4	0	0	4	56
Course Category	DSC					
Course Objectives: This course is designed to introduce the process of designing data structures to solve fairly complex problems. It imparts the following. 1. Basic concepts of data structures and algorithms 2. Concept and use of lists, trees, heaps and graphs 3. Searching techniques 4. Enable students to write algorithms for solving problems with the help of fundamental data structures						
Course Outcomes: Develop the skill set to 1. Choose the appropriate data structure for solving a specific problem 2. Analyze the algorithms to determine the time and computation complexity 3. Implement different search techniques 4. Solve multiple problems using lists, trees, heaps and graphs 5. Determine the best suited data structure to solve the problems						
Prerequisites for the Course: UCSH-101						
Unit	Topic	Contents				Periods
1	Fundamentals and Linear Lists	Abstract data type, Algorithm Analysis using Big-O notation, Stack,, Queue, Linked List, Circularly linked list, Doubly linked list, Applications				10
2	Trees	Binary Tree, General Tree, Binary Search Tree, AVL Search Tree, Applications				14
3	Heaps	Basic concepts, Operations, Algorithms, Implementation, Applications				8
4	Graph	Basic concepts, Representations, Traversals, Algorithms, Applications				14

5	Searching	Linear Search and variations, Binary Search, Hashed list search, Collision resolution	10
		Total	56
Key Text: Richard F. Gilberg and Behrouz A. Forouzan, Data Structures - A Pseudocode approach with C, second edition, Cengage Learning, 2005. Chapters : 3 – 9, 11, 13 References: Mark Allen Weiss, Data Structures and Algorithm Analysis in C, Second Edition, Pearson Education, 1997.			

Course Title	Software Lab in C - Part II	L	T	P	C	Periods
Course Code	UCSH-202	0	0	6	3	84
Course Category	DSC					
Course Objectives: This lab course introduces the basics of data structures using C language. It would enable a student to design & implement data structures in C to solve fairly complex problems thereby improving data structure design skills.						
Course Outcomes: Develop the skill set to 1. Program basic data structures in C. 2. Implement Lists, Trees, Heaps and Graphs appropriately to solve problems 3. Implement different searching techniques						
Prerequisites for the Course: UCSH-102						
Unit	Topic	Contents				Periods
1	Pointers	Simple programs for learning C Pointer Concepts, Pointers and arrays, passing pointers in functions				6
2	Linked Lists	Implementing Singly linked list, doubly linked list, circular linked list with a few applications				18
3	Trees	Implementing General Tree, Binary Tree, Binary Search Tree, AVL Trees with a few applications				24

4	Heaps and Graphs	Implementing Heaps, Adjacency list representation of graphs, breadth first traversal, depth first traversal, basic functions in graph	24
5	Search	Linear search, Binary search, hashed list search	12
		Total	84

References:

1. Computer Science - A Structured Programming Approach using C by Behrouz Forouzan & Richard Gilberg.
2. The C Programming Language by Dennis Ritchie
3. Yashwant Kanetkar, Pointers in C.

Semester-III

Course Title	Computer Organization and Design	L	T	P	C	Periods
Course Code	UCSH-301	4	0	0	4	56
Course Category	DSC					
Course Objectives: This course introduces students to the fundamental concepts of computing systems. This is to familiarize students to equip them with an understanding of hardware design including logic design, basic structure, and behavior of the essential functional modules of the computer. It also gives them how the system interacts to accomplish the required processing for the needs of the user.						
Course Outcomes: Develop the skill set to: 1. To know the basics of Instruction Set Architecture (ISA) and organization of Computer components. 2. Solve simple boolean algebra problems as related to designing computer logic, through combinational and sequential logic circuits. 3. Understand how computers represent and manipulate data and convert it between different number systems. 4. Understand the memory hierarchies and processor structure and functions						
Prerequisites for the Course: NA						
Unit	Topic	Content				Period
1	Arithmetic and Logic	Basics: Functional Blocks in a Computer System, Number system and Computer Arithmetic, Boolean algebra, Gates and combinational logic, Sequential Logic circuits.				20
2	Central Processing Unit	Addressing Modes, Instruction Set and Instruction Execution Flow, processor structure and function				12
3	Control Unit	Hardware and Micro-program based control Unit Design				10
4	Memory	Memory Architecture,Cache,Internal memory				12
		Total				56 Hrs

Key Text:

1. William Stallings, Computer organization and architecture: designing for performance, Pearson Education India, 2000.

Chapters : 3 (3.1-3.2), 4 (4.1-4.3), 5 (5.1), 7 (7.1-7.6), 9 (9.1-9.5), 10 (10.1-10.2), 11 (11.1), 12 (12.1 to 12.4), 20 (20.1-20.5)

Suggested Reading:

1. Carl Hamacher, Zvonko Vranesic and Safwat Zaky. "Computer Organization", McGraw Hill, 2011.

2. David A. Patterson, John L. Hennessy. "Computer organization and design: the hardware/software interface". Elsevier, 2011.

Course Title	Discrete Mathematics for Computer Science	L	T	P	C	Periods
Course Code	UCSH-302	4	0	0	4	56
Course Category	DSC					
Course Objectives: It emphasizes conceptual understanding, quantitative reasoning, and contemporary applications by maintaining a dynamic balance among theory, applications, modeling, and drill. Students work to develop a deeper understanding of a range of mathematical topics using interactive math tutorials that allow the student to solve math problems interactively. Students will gain some confidence on how to deal with problems which may arrive in computer science in the near future.						
Course Outcomes: Develop the skill set to 1. Write and interpret mathematical notation and mathematical definitions 2. Improve the proof writing skills 3. Understand the basic mathematical structures, counting principles 4. Solve linear recurrence relations 5. Describe and solve some real time problems using concepts of graph theory						
Prerequisites for the Course: NA						
Unit	Topic	Contents				Periods
1	Logic	Propositional Logic (PL), Applications of PL, Propositional Equivalences , Predicates and Quantifiers, Nested Quantifiers, Rules of Inference; Introductions to Proofs, Proof Methods and Strategy				12

2	Counting Principles	The Basics of Counting, The Pigeonhole Principle, Permutations and Combinations, Binomial Coefficients and Identities, Generalized Permutations and Combinations, Applications of Recurrence Relations, Solving Linear Recurrence Relations	12
3	Relations	Relations and Their Properties, Representing Relations, Closures of Relations, Equivalence Relations, Partial Orderings	12
4	Graphs and Trees	Graphs: Graph Models, Special Types of Graphs, Representing Graphs and Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Shortest-Path Problems, Planar Graphs, Graph Coloring Trees: Introduction to Trees, Applications of Trees, Tree Traversal, Spanning Trees, Minimum Spanning Trees	12
5	Algebraic Structure	Algebraic Systems: Example and general Properties, Semigroups and Monoids: Definitions and Examples, Groups: Subgroups and Homomorphisms	8
		Total	56

Key Text:

1. Kenneth H. Rosen, Discrete mathematics and its applications, Eight Edition, McGraw Hill, 2019
2. Tremblay J.P and Manohar R, Discrete Mathematical Structures with Applications to Computer Science, TMH Pub. Co. Ltd, 2010.

Suggested Reading:

1. Kolman, Busby, Ross: Discrete Mathematical Structures, Pearson education, 4th Edition.
2. T. Veerarajan: Discrete Mathematics, with Graph Theory and Combinations(2008), Tata McGraw – Hill Publishing Company Ltd, New Delhi
3. Discrete Mathematics, Dr A Singaravelu et al, Meenakshi Agency.

Semester-IV

Course Title	Database Management Systems	L	T	P	C	Periods
Course Code	UCSH-401	4	0	0	4	56
Course Category	DSC					
Course Objectives: To understand different issues involved in the design and implementation of database systems.						
Course Outcomes: Develop skill set to 1. Draw ER diagrams for a given problem 2. Design relational database with normalization 3. Handle concurrency control and storage requirements of a database						
Prerequisites for the Course: UCSH-202,UCSH-301						
Unit	Topic	Contents				Periods
1	Introduction	Characteristics of database approach, Purpose of Database Systems-Drawbacks of file System, data models, DBMS architecture and data independence, E-R Modeling: Entity Course Categories, entity set, attribute and key, relationships, relation Course Categories, roles and structural constraints, weak entities, enhanced E-R and object modeling, Sub Classes:, Super classes, inheritance, specialization and generalization				14
2	File Organization	Indexed sequential access files, implementation using B++ trees, hashing, hashing functions, collision resolution, extendible hashing, dynamic hashing approach implementation and performance.				14
3	Relational Data Model	Relational model concepts, relational constraints, relational algebra. SQL: SQL queries, programming using SQL EER and ER to relational Mapping: Database design using EER to relational language, The Unified Modeling language UML-features of Good Relational Design.				14

4	Data Normalization	Functional dependencies, Normal form up to 3rd normal form. Concurrency Control: Transaction processing, locking techniques and associated, database recovery, security and authorization. Recovery Techniques, Database Security	14
		Total	56 Hr
Key Text: 1. R. Elmasri and SB Navathe, Fundamentals of Database Systems, Fourth Edition, Addison Wesley, 2004 Chapters: 1.1, 1.2, 1.3, 2.1, 2.2, 3.3, 3.4, 3.5, 4.1, 4.2, 5.1, 5.2, 6.2, 7.1, 8.4, 9.1, 10.2, 10.3, 10.4, 13.8, 14.3, 18.1, 18.2, 19.1, 23.1			
Suggested Readings: 1. Abraham Silberschatz, Henry Korth, S. Sudarshan, Database Systems Concepts, 4th Edition, McGraw Hill, 1997. 2. Jim Melton, Alan Simon, Understanding the new SQL: A complete Guide, Morgan Kaufmann Publishers, 1993.			

Course Title	Object Oriented Programming Concepts	L	T	P	C	Periods
Course Code	UCSH-402	4	0	0	4	56
Course Category	DSC					
Course Objectives: To understand and implement standard tools and techniques for software development using object oriented approach.						
Course Outcomes: Develop the skill set to 1. Analyze and design real world problems using OOPS concepts viz: Encapsulation, object identity, inheritance and polymorphism. 2. Handle exceptions and write generic programming 3. Write simple C++ programs using basic object oriented paradigms						
Prerequisites for the Course: UCSH-102						

Unit	Topic	Content	Periods
1	Introduction	Introducing Object-Oriented Approach, complexity of the structure and design of complex systems, Relating to other paradigms (functional, data decomposition). Basic terms and ideas: Abstraction, Encapsulation, Inheritance, Polymorphism, Review of C, Difference between C and C++ - cin, cout, new, delete operators.	14
2	Classes and Objects	Encapsulation, information hiding, abstract data types, Object & classes, attributes, methods, C++ class declaration, State identity and behavior of an object, Constructors and destructors, instantiation of objects, Default parameter value, objects, dynamic memory allocation, Metaclass/abstract classes.	14
3	Inheritance and Polymorphism	Inheritance, Class hierarchy, derivation – public, private & protected, Aggregation, composition versus classification hierarchies, Polymorphism, Categorization of polymorphism techniques, Method polymorphism, Polymorphism by parameter, Operator overloading, Parametric polymorphism	14
4	Generic function	Template function, function name overloading, Overriding inheritance methods, Runtime polymorphism, Multiple Inheritance. Files and Exception Handling: Persistent objects, Streams and files, Namespaces, Exception handling, Generic Classes	14
		Total	56
Key Text: 1. A.R.Venugopal, Rajkumar, T. Ravishanker, Mastering C++, TMH, 1997. Chapters: 1 (1.1.-1.18), 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, 15, 16 (16.1-16.3), 17 (17.1-17.2), 18 (18.1-18.9), 19 (19.1-19.7)			

Suggested Readings:

1. R. Lafore, "Object Oriented Programming using C++", Galgotia Publications, 2004.
2. James Rumbaugh "Object Oriented Modeling and Design" Prentice Hall Publications, 1991
3. D. Parsons, "Object Oriented Programming with C++", BPB Publication.
4. Steven C. Lawlor, "The Art of Programming Computer Science with C++", Vikas Publication.
5. Schildt Herbert, "C++: The Complete Reference", 4th Ed., Tata McGraw Hill, 1999.
6. Tony Gaddis, Watters, Muganda, "Object -Oriented Programming in C++", 3rd Ed., Wiley Dreamtech, 2004
7. Grady Booch Object oriented Analysis and Design with applications
8. S. B. Lippman & J. Lajoie, "C++ Primer", 3rd Edition, Addison Wesley, 2000.

Course Title	Software Lab in C++	L	T	P	C	Periods
Course Code	UCSH-403	0	0	6	3	84
Course Category	DSC					
Course Objectives: To analyze a real world problem and code using object oriented concepts of C++ .						
Course Outcomes: Develop the skill set to 1. Understand and implement object oriented paradigms viz., Encapsulation, object identity, inheritance and polymorphism. 2. Develop a mini project.						
Prerequisites for the Course: UCSH-102, UCSH-202						
Unit	Topic	Contents				Periods
1	Classes and Objects	To set and get the values given by the user, calculate the area/perimeter of geometrical shapes, Using friend functions. Static members and functions to count the number of objects created. Using constructors (default, parameterized, overloaded) defined in (local, global and function) and destructors along with their order of invocation in addition / multiplication of complex numbers (passing objects as parameters)by using inline functions. matrix multiplication using dynamic				22

		objects.	
2	Inheritance: Single, Multiple, Hierarchical, Multilevel, Hybrid.	To write sample codes with different access specifiers, visibility modes and various types of inheritance, To use constructors and destructors in all the types of inheritance to understand the order of invocation.	12
3	Polymorphism, Templates, Exception handling, Namespaces, Files	Operator overloading on: arithmetic, logical, assignment, relational, unary, subscripting, new and delete operators, using abstract classes with virtual functions. Developing sample codes for swapping, maximizing two numbers using templates. Handling division by zero exception, creating own namespace. To read and write into a file.	22
4	Mini project	Student academic/ non-academic activities, Banking systems/ Self-reliance departments/ etc., using all the concepts.	28
		Total	84
Reference: 1. A.R. Venugopal, Rajkumar, T. Ravishanker “Mastering C++”, TMH, 1997. 2. Schildt Herbert, “C++: The Complete Reference”, 4th Ed., Tata McGraw Hill, 1999.			

**SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(DEEMED TO BE UNIVERSITY)**



**Syllabus for Minor Disciplines offered by
Department of Mathematics and Computer Science**

PRASANATHI NILAYAM- 515134
Sri Sathya Sai District, Andhra Pradesh
Tel: 08555- 287239
www.sssihl.edu.in

(EFFECTIVE FROM JUNE 2024-2025 ONWARDS)

SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

Syllabus for Minor Disciplines Offered for Undergraduate Programs
(Effective from 2024-2025 onwards)

INTRODUCTION

Department of Mathematics and Computer Science (DMACS) offers minor courses in the following Disciplines for undergraduate students with required eligibility:

1. Mathematics
2. Actuarial Science
3. Data Science

These Minor degrees or courses will be offered subject to available logistics such as faculty, timetable slots etc. Proper understanding must be sought from the course coordinator/HoD before choosing to do a minor course in a particular semester.

Minor Certification: A candidate who chooses to do a minor has to earn a minimum of 12 credits in 3 years of B.S program or 16 credits in 4 years of B.S Hons/B.S Hons with Research program. It is to be noted that, the minimum credits mentioned above must be earned from the courses which are of difficulty level 300 or above. A student majoring in a discipline cannot obtain a minor in the same discipline.

MOOCs Courses for Minor Credit Transfer: A candidate can choose to earn up to 12 minor credits from online MOOCs Courses.

1. Mathematics Minor

Prerequisite: 12th standard from any stream with Mathematics as one of the core subjects
The following courses are offered for Mathematics as minor.

100 Differential Calculus
100 Integral Calculus
200 Linear Algebra -I/300 Complex Analysis
200 Algebraic Structures I/300 Methods of Differential Equations
300 Optimization Techniques
300 Numerical Analysis
400 Algebraic Structures -II
400 Mathematical Modelling

Note: For detailed syllabus, consult the syllabus document for the program BSc Mathematics/BSc(Hons) Mathematics/BSc(Hons) Mathematics with Research.

Y. Lakshmi

2. Actuarial Science Minor

Prerequisite: 12th standard from any stream with Mathematics as one of the core subjects

The following list of courses offered for Computer Science as minor

- 101 Actuarial Probability and Statistics
- 102 Actuarial Mathematics
- 301 Actuarial Applied Statistical Methods
- 302 Business Economics with Actuarial Applications
- 303 Business Accounting & Finance with Actuarial Applications
- 401 Regulation and Financial reporting with Actuarial Application

Note: For a detailed syllabus, consult the syllabus document for the program BSc Mathematics/BSc(Hons) Mathematics/BSc(Hons) Mathematics with Research.

3. Data Science Minor

Prerequisite: 12th standard from any stream with Mathematics as one of the core subjects

The following list of courses offered for Data Science as minor

- UDSC-101: Calculus
- UDSC-201: Probability and Statistics
- UDSC-301: Statistics for Data Science
- UDSC-401: Linear Algebra for Data Science
- UDSC-402: Data Analysis and Visualization
- UDSC-501: Optimization for Machine Learning
- UDSC-502: Data Mining and Machine Learning
- UDSC-601: Big Data Analytics
- UDSC-701: Cloud Computing

Data Science Minor Courses Syllabus

Course Title	CALCULUS	L	T	P	C	Periods
Course Code	UDSC-101	3	1	0	4	56
Course Category	IDM					
Course Objectives: Calculus provides the tools and methods for analyzing and solving problems of change and motion quantitatively. It introduces mathematical operations of differentiation and integration and develops concepts of central importance in mathematical sciences.						
Course Outcomes: 1. Develop the skill set to compute the slope of tangent at a point for given curve 2. write a differentiable function into a series 3. find the maxima and minima of function in both one and two variables 4. compute the area under the curve for a given surface 5. perform different operators on vectors like gradient, divergence and curl 6. compute the volume of a given solid.						
Prerequisites for the Course: 10 + 2 Mathematics						
Unit	Topic	Contents				Periods
1	Basics	Sets, Real valued functions, Graphs of functions, Limits, Continuous and differentiable functions. Sequences-Limit, convergence, divergence, oscillation. Series - Convergence, divergence, oscillation, Limit points, boundary points, interior points, open set, closed set.				6
2	Differential Calculus	Successive differentiation, Leibnitz Theorem, Partial Derivatives, Euler's theorem, change of variables, total differentiation, Errors and approximation. Taylor’s series in two variables. Maxima and Minima of two or more variables, Jacobians and properties - Lagrange’s method of undetermined multipliers.				15
3	Integral Calculus	Definite integral, Multiple integrals. Change of order of Integration. Transformation of integral from Cartesian to polar. Applications in areas, volume and surfaces. Beta function, Gamma function, Relation between beta and gamma function.				15
4	Vector Algebra	Equation of plane, Straight line, Intersection of lines, Shortest distance between two lines, Intersection of planes, Sphere, Equation of tangent plane, Cone, Cylinder, Quadratic surfaces.				14
5	Vector Calculus	Differentiation of vectors, Curves in space, Vector operator ∇ , Gradient, Divergence and Curl.				6
		Total				56

Key Text: Dr. B.S.Grewal, Higher Engineering Mathematics, 44th Edition, Khanna Publications, 2021, Chapters 3(3.11 to 3.21), 4(4.1 to 4.2) , 5 (5.1 to 5.12), 6 (6.8) 7 (7.1 to 7.8, 7.14 to 7.16) 8 (8.1 to 8.2, 8.4 to 8.9) References: Calculus, George B. Thomas, et al, 12th Edition, 2009	
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Course Title	Probability and Statistics	L	T	P	C	Periods
Course Code	UDSC-201	4	0	0	4	56
Course Category	IDM					
Course Objectives: Probability and Statistics is the branch of mathematics that deals with modeling uncertainty. It is important because of its direct application in areas such as computer science, finance, and telecommunications. It forms the fundamental basis for many other areas in the mathematical sciences including modern optimization methods and risk modeling. This course provides students with a formal treatment of probability and statistics.						
Course Outcomes 1. Develop the skill set to solve real world problems based on axioms of probability, conditional probability and Bayes’ theorem 2. understand the several discrete and continuous random variables 3. compute covariance and correlation of two or more than two dimensional data identify statistical distributions by observing real data sets 4. compute the moment-generating function for a given random variable 5. plot histograms, Box-plot for a given univariate data 6. plot scatter diagram for a bi-variate data						
Prerequisites for the Course: This course can be taken after UDSC-101 Calculus						
Unit	Topic	Contents				Periods
1	Elements of Probability	Sample Space, Events, counting Techniques, Axioms of Probability, Addition Rules, Multiplication Rules, Conditional Probability, Independence, Bayes’ Formula and Random Variables				12
2	Discrete Random Variables	Probability Distributions and Probability Mass Functions, Cumulative Distribution Functions, Mean and Variance of a Discrete Random variable Special Discrete Random variables: Uniform, Binomial, Geometric, Negative Binomial, Hypergeometric and Poisson Distributions				12

3	Continuous Random Variables	Probability Distributions and Probability Mass Functions, Cumulative Distribution Functions, Mean and Variance of a Discrete Random variable Special Continuous Random variables: Uniform, Exponential, Normal, Erlang, Gamma, Weibull, Lognormal and Beta Distributions. Normal Approximation to the Binomial and Poisson Distributions	14
4	Joint probability Distributions	Joint distribution of Two and more than two Random Variables, Conditional Probability Distributions and Independence, Covariance and Correlation, Common Joint Distributions, Linear Functions and General Functions of Random Variables, Moment-Generating Functions	12
5	Descriptive Statistics	Numerical Summaries of Data, Stem-and-Leaf Diagrams, Frequency Distributions and Histograms, Box Plots, Scatter Diagrams, Probability Plots.	6
		Total	56

Key Text:

Montgomery Douglas C., Runger George C., Applied Statistics and Probability for Engineers, Seventh Edition, Wiley, 2018, Chapters: 2 - 6.

Suggested Readings:

1. Sheldon M Ross, Introduction to Probability and Statistics for Engineers and Scientists, 6th Edition, Elsevier Academic Press, 2020
2. Irwin Miller Marylees Miller, John E. Freund's Mathematical Statistics with Applications, 8th Edition, Pearson Publications

Course Title	Statistics for Data Science	L	T	P	C	Periods
Course Code	UDSC-301	3	1	0	4	56
Course Category	IDM					
Course Objectives: The main purpose of Statistical Inference is to make an inference about a population based on information contained in a random sample selected from that population, and to provide an associated measure of goodness for the inference. Another purpose is hypothesis testing for mean and variance of a normal distribution.						
Course Outcomes: 1. Develop the skill set to Estimate both point and interval estimations for a given parameters based on a sample from a normal population 2. Analyze the estimated parameter 3. Find an estimator for population proportion based on sample data 4. Develop testing hypothesis for a given real world problem 5. Apply different statistical tests based on data 6. Model the relationship between a scalar response and one or more explanatory variables 7. Find an appropriate regression model for a given data						
Prerequisites for the Course: Probability and Statistics, Calculus						
Unit	Topic	Contents				Periods
1	Point Estimation	Sampling Distributions and the Central Limit Theorem, Point Estimation: Unbiased Estimators, Variance of a Point Estimator, and Mean squared Error of an Estimator, and Methods of Point Estimation: Method of Moments, Maximum Likelihood Estimators, Bayesian Estimation of Parameters;				10
2	Interval Estimation for a Single Sample	Confidence Intervals on mean of a Normal Distribution, Confidence Interval on variance of a Normal Distribution, Large Sample Confidence Interval for a Population Proportion, Bootstrap Confidence Interval, Tolerance and Prediction Intervals				9
3	Testing Hypotheses for a Single Sample	Hypothesis Testing, Tests on the Mean and variance of a Normal Distribution, Tests on a Population Proportion, Testing for Goodness of Fit and Contingency Table Tests				10
4	Statistical Inference for Two samples	Inference on the Difference in Means of two Normal Distributions, A Nonparametric Test for the Difference in Two Means, Paired t-test, Inference on the variance				12

		of Two Normal Distributions, Inference on Two Population Proportions	
5	Simple Linear Regression	Simple Linear Regression, Least squares estimators, Hypothesis tests and Confidence Intervals in Single Linear Regression, Adequacy of the Regression Model: Residual Analysis, Coefficient of Determination, Regression on Transformed Variables, Logistic Regression, Polynomial Regression Models	15
		Total	56
Key Text : Montgomery Douglas C., Runger George C., Applied Statistics and Probability for Engineers, Seventh Edition, Wiley, 2018 Chapters: 7 -9 (up to 9.8), 10, 11 Suggested Readings: 1. Sheldon M Ross, Introduction to Probability and Statistics for Engineers and Scientists, Sixth Edition, Elsevier Academic Press, 2020 2. Irwin Miller, Marylees Miller, John E. Freund's Mathematical Statistics with Applications, Eighth Edition, Pearson Publications			

Course Title	Linear Algebra for Data Science	L	T	P	C	Periods
Course Code	UDSC-401	3	0	1	4	70
Course Category	IDM					
Course Objectives: This course provides an introduction to Linear algebra and its applications to other disciplines.						
Course Outcomes: By the end of the course, students will be able to: 1. Understand the use of linear algebra in modeling concepts. 2. Perform matrix calculations effectively. 3. Describe the concepts of vector spaces and their subspaces through various examples. 4. Differentiate between linearly independent and dependent sets and determine the basis for finite-dimensional vector spaces. 5. Analyze linear transformations between vector spaces, discern their properties using the image and kernel of the transformation. 6. Express linear transformations in matrix form and discuss their properties. 7. Understand the use of SVD. 8. Explain characteristic polynomials and comprehend the concepts of eigenvalues and eigenvectors in the context of endomorphisms.						
Prerequisites for the Course: 10+2 Mathematics, UDSC-101						

Unit	Topic	Contents	Periods.
1	Vector Spaces	Vectors - Linear Combinations, Lengths, Dot products. Matrices-Linear Equations, Elimination method, Rules of Matrix operations, Inverse, LU factorization, Transposes and permutations.	8
2	Fundamental Spaces	Vector space-Spaces of vectors, Null space, Complete solution to nonhomogeneous equation, Independence, Basis, Dimension. Subspaces - Dimension of Four subspaces, Orthogonality, Projection, Orthonormal bases, Gram Schmidt orthogonalisation	8
3	Determinants	Determinants-Properties, Permutations and Cofactors, Cramer's rule, Inverse, Volume. Eigenvalues and Eigenvectors - Eigenvalues, Diagonalisation, System of differential equations, Symmetric matrices, Positive Definite matrices	8
4	SVD	Singular Value Decomposition - Image processing, Bases and matrices, Principal Component Analysis, Geometry of SVD	10
5	Linear Transformations	Linear Transformations - Introduction, Matrix of Linear Transformation, Search for a good basis	8
		Total	42
Key Text Gilbert Strang, Introduction to Linear Algebra, 5th Edition, Wellesley-Cambridge Press, 2016 Chapters: 1 (1.1-1.3), 2 (2.1-2.7), 3 (3.1-3.5), 4 (4.1-4.4), 5 (5.1-5.3), 6 (6.1-6.5), 7 (7.1-7.4) https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/video_galleries/video-lectures/ Retrieved on Jan 31, 2023			
REFERENCES 1. Stephen Boyd & Lieven Vandenberghe, Introduction to Applied Linear Algebra - Vectors, Matrices and Least Squares, Cambridge University Press, 2018 2. Philip N. Klein, Coding the Matrix - Linear Algebra through Applications to Computer Science, Newtonian Press, 2013			

UDSC-401: Practicals: Linear Algebra for Data Science			
Unit	Topic	Content	Periods
1	Python for Linear Algebra	The following programs are to be implemented in basic python - Dot Product of Vectors, Matrix Multiplication with basic python - Determinant of a Matrix, Matrix Inverse with basic python - Rank of a Matrix with basic python - Gaussian Elimination with basic python	12
2	Numpy and its usage in Linear Algebra	The following are to be implemented using numpy - Eigenvalues, Eigenvectors, SVD - Eigenfaces - Linear Regression - Matrix Exponential	16
		Total	28
Key Text: Leo Chin, Tanmay Dutta, NumPy Essentials, Packt Publishing, 2016			
REFERENCES			
Umit Mert Cakmak, Mert Cuhadaroglu, Mastering numerical computing with Numpy, Packt Publishing, 2018			

Course Title	Optimization for Machine Learning	L	T	P	C	Periods
Course Code	UDSC-501	3	0	1	4	70
Course Category	IDM					
Course Objective: To modeling and discussion of documented real-world applications. Study of mathematical programming algorithms for optimization. Apply the mathematical results and numerical techniques of optimization theory to concrete Engineering problems.						

Course Outcome: Develop the skill set to 1. discovery, study and solve optimization problems; 2. Investigate, study, develop, organize and promote innovative solutions for various applications. 3. Apply mathematical skills to model optimization problems. 4. Critically analyze theoretical principles and choose the relevant optimization techniques for a specific problem. 5. Exposed to solutions for various types of optimization problems and will be enabled to adopt them for the situation at hand.			
Prerequisites for the Course: 10+2 Mathematics, Python programming, UDSC-101 Calculus			
Unit	Topic	Description	Periods
1	Introduction	Mathematical Review – Lines – Hyperplanes and linear varieties – convex sets – neighborhoods – polytopes and polyhedra – derivative matrix – level sets and gradients-Taylor’s series	6
2	Unconstrained Optimization	Unconstrained Optimization – conditions for local minimizers – one-dimensional search methods including golden section, Fibonacci, bisections and Newton’s method – Gradient methods including steepest descent methods, Newton's method and its analysis.	8
3	Linear Programing	Modeling with Linear Programming The Simplex method and Duality	8
4	Non-Linear Constrained Optimization	Non-linear constrained optimization – problems with equality constraints – Lagrange condition – second-order condition – minimizing quadratics subject to equality constraints – problems with inequality constraints – Karush-Kuhn-Tucker conditions – second-order conditions.	8
5	Convex Optimization	Convex Optimization Problems - Convex Functions - Convex Optimization Problems	6
6	Machine Learning view for Optimization	The minutiae of Gradient Descent, Properties of Optimization in ML, Computing Gradient wrt vectors, Linear Regression and Optimization models for Binary Targets	6
		Total	42

Suggested/Additional Readings:

1. L.R. Foulds, Optimization Techniques, Springer, Utm, 1981
2. <http://freevideolectures.com/Course/3072/Numerical-Optimization>
3. Hamdy A. Taha, Operations Research, Pearson Publication, 9th edition, (2014).
4. Boyd, Stephen, and Lieven Vanderberghe, Convex Optimization. Cambridge, UK: Cambridge University Press, 2004.

UDSC-501: Practicals - Optimization for Machine Learning

Unit	Topics	Contents	Periods
1	Unconstrained Optimization	Implementation of Unconstrained Optimization using Python: Newton-Raphson method; Implementation of Gradient Ascent and Gradient Descent methods	10
2	Solving LPP	Solving LP problems using Excel solver; Sensitivity Analysis using Excel solver; Solving LP problems using <i>scipy.optimize.linprog</i>	10
3	Python Library <i>scipy.optimize</i>	Univariate function minimization; Constrained and Unconstrained minimization of multivariate scalar function; Global Optimization; Least-Square Optimization; Custom minimizers	8
		Total	28

References:

1. SciPy Reference Guide, Retrieved July 7, 2022 from <https://docs.scipy.org/doc/scipy-1.7.1/scipy-ref-1.7.1.pdf>

Section 2.4 (Optimization - *scipy.optimize*)
2. Christian Hill, Learning Scientific Programming with Python, Second Edition, Cambridge University Press, 2020
3. Edwin K. P. Chong and Stanislaw H. Zak, An Introduction to Optimization, Fourth Edition, Wiley Inter - Science Publication, John Wiley and Sons Inc., 2013.
4. Hamdy A.Taha, Operations Research- An Introduction, 10th Edition, Pearson Education - 2017

Course Title	Data Mining and Machine Learning	L	T	P	C	Periods
Course Code	UDSC-502	3	0	1	4	70
Course Category	IDM					
Course Objectives: 1. To introduce students to the basic concepts and techniques of Machine Learning. 2. To become familiar with regression methods, classification methods, Pattern Mining and clustering methods. 3. To become familiar with Kernel Methods and Dimensionality reduction Techniques						
Course Outcomes: Develop the skill set to 1. Gain knowledge about basic concepts of Machine Learning 2. Identify machine learning techniques suitable for a given problem 3. Solve the problems using various machine learning techniques 4. Apply Dimensionality reduction techniques						
Prerequisites: UDSC-101/UMAT-101, UDSC-102, UDSC-201, UDSC-203 or Equivalent						
Unit	Title	Contents			Periods	
1	Introduction	Data Matrix, Numeric Attributes, Categorical Attributes, Graph Data, Kernel Methods, High-dimensional Data, Dimensionality Reduction, Types of Machine Learning			8	
2	Regression	Linear Regression, Logistic Regression, Neural Networks, Regression Evaluation			8	
3	Classification	Probabilistic Classification, Decision Tree Classifier, Linear Discriminant Analysis, Support Vector Machines, Classification Assessment.			12	
4	Clustering	Representative-based Clustering, Hierarchical Clustering , Density-based Clustering, Spectral and Graph Clustering, Clustering Validation			14	

5	Frequent pattern mining	Itemset Mining, Summarizing Itemsets Sequence Mining, Graph Pattern Mining.	14
Total 56			
Key Text(s): 1. Mohammed J. Zaki, Wagner Meira, Jr., Data Mining and Machine Learning: Fundamental Concepts and Algorithms, 2nd Edition, Cambridge University Press, March 2020. 2. Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman, The Elements of Statistical Learning, Second Edition, Springer, 2009 Retrieved on July 05, 2022 from https://hastie.su.domains/Papers/ESLII.pdf 4. Christopher M Bishop, Pattern Recognition and Machine Learning, Springer, 2006 5. Rogers and Girolami, A First Course in Machine Learning, Chapman and Hall/CRC, 2015			
References: 1. Barber, Bayesian Reasoning and Machine Learning, Cambridge University Press, 2012 2. Hal Daumé III, A Course in Machine Learning, e-Edition, 2012 3. Mitchell, Machine Learning, McGraw Hill, 1997 4. CS229-Machine Learning, Stanford University, Retrieved June 30, 2022 from https://see.stanford.edu/Course/CS229			
UDSC-502: Practicals- Data Mining and Machine Learning			
Unit	Title	Contents	Periods
1	Introduction	Review numpy, pandas and matplotlib for statistical, Probabilistic, algebraic and geometric views of data matrix Implementation of Linear algorithms, Non-Linear Algorithms and ensembling algorithms, bias-variance tradeoff, from scratch in python	12
2	Python Library scikit-learn	Develop different ML algorithms using scikit-learn	16
Key Text(s): 1. Jason Brownlee, Master Machine Learning Algorithms, Discover How They Work and Implement them from Scratch, Machine Learning Mastery, eBook, 2017 Retrieved June 30, 2022 https://github.com/AmandaZou/master-machine-learning-algorithms 2. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'reilly, 2019 3. Scikit-learn Documentation, retrieved June 30, 2022 from scikit-learn Tutorials — scikit-learn 1.0.2 documentation			

Course Title	Big Data Analytics	L	T	P	C	Periods
Course Code	UDSC - 601	4	0	0	4	56
Course Category	IDM					
Course Objectives: 						

6	Social Network Graphs	Clustering, Discovering of communities, Partitioning, Finding overlapping communities, Simrank, Counting Triangles, Neighborhood properties	12
		Total	56
Key Text(s): Anand Rajaraman, Mining Massive Datasets, Stanford University Press, 2014 Chapters: 2.1 - 2.3, 3.1 - 3.7, 4.1 - 4.6, 5.1 - 5.4, 6.1 - 6.4, 10			

Course Title	Cloud Computing	L	T	P	C	Periods
Course Code	UDSC-701	4	4	0	4	56
Course Category	IDM					

Course objectives:

This course gives students an insight into the basics of cloud computing along with virtualization, cloud computing is one of the fastest growing domains from a while now. It will provide the students basic understanding about cloud and virtualization along with how one can migrate over it.

Course outcomes: Develop the skill to

1. analyze the cloud computing setup with its vulnerabilities and applications using different architectures.
2. design different workflows according to requirements and apply map reduce programming model.
3. apply and design suitable virtualization concepts, cloud resource management and design scheduling algorithms.
4. create combinatorial auctions for cloud resources and design scheduling algorithms for computing clouds
5. assess cloud storage and security, the risks involved, its impact and develop cloud applications
6. broadly educate to know the impact of engineering on legal and social issues involved in addressing the security issues of cloud computing.

Prerequisites: Courses on Operating Systems, Computer Networks

Unit	Title	Contents	Periods
1	Cloud Computing Overview	Origins of Cloud computing – Cloud components - Essential characteristics – On-demand self service, Broad network access,	3

		Location independent resource pooling ,Rapid elasticity , Measured service, Comparing cloud providers with traditional IT service providers, Roots of cloud computing.	
2	Cloud Insights	Architectural influences – High-performance computing, Utility and Enterprise grid computing, Cloud scenarios – Benefits: scalability ,simplicity ,vendors ,security, Limitations – Sensitive information - Application development- security level of third party - security benefits, Regularity issues: Government policies.	3
3	Cloud Architecture - layers and Models	Layers in cloud architecture, Software as a Service (SaaS), features of SaaS and benefits, Platform as a Service (PaaS), features of PaaS and benefits, Infrastructure as a Service (IaaS), features of IaaS and benefits, Service providers, challenges and risks in cloud adoption. Cloud deployment model: Public clouds	10
4	Cloud Simulators CloudSim and Green Cloud	Introduction to Simulator, understanding CloudSim simulator, CloudSim Architecture(User code, CloudSim, GridSim, SimJava) Understanding Working platform for CloudSim, Introduction to GreenCloud	20
5	Introduction to VMWare Simulator	Basics of VMWare, advantages of VMware virtualization, using Vmware workstation, creating virtual machines-understanding virtual machines, creating a new virtual machine on local host, cloning virtual machines, virtualizing a physical machine, starting and stopping a virtual machine.	20
Total			56
Key Text: 1. Cloud computing a practical approach - Anthony T.Velte , Toby J. Velte Robert Elsenpeter, TATA McGraw- Hill , New Delhi – 2010 2. Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online - Michael Miller - Que 2008			
Reference Texts: 1. Cloud computing for dummies- Judith Hurwitz , Robin Bloor , Marcia Kaufman ,Fern Halper, Wiley Publishing, Inc, 2010 2. Cloud Computing (Principles and Paradigms), Edited by Rajkumar Buyya, James Broberg, Andrzej Goscinski, John Wiley & Sons, Inc. 2011			

Skill Enhancement Courses

PAPER CODE: UCSH-103		Software Lab in Python-I	CREDITS: 1 (0-0-2)
Course Category:		SEC	Total Periods: 28
Prerequisite (If any): NA			
Course Objectives: This course provides an introduction to the Python programming language. It covers the fundamentals of programming, including variables, data types, control structures, and functions. Students will learn how to write Python programs to solve simple problems and develop basic programming skills.			
Unit	Description		Periods
I	Introduction to Programming using Python Introduction to Programming Overview of the Python programming language Setting up the development environment Running Python program		2
II	Data Types and Operators int, float, complex, str, list, tuple, bytes, byte array, dictionaries Boolean type, Arithmetic operator, Relational operator, Logical or Boolean operator, Assignment Operator, Bitwise operator		8
III	Control Structures Conditionals: if, else, elif statements Loops: for and while loops Loop control statements: break, continue		8
IV	Functions Python built in functions, Python user defined functions, Python packages functions; Defining and calling Function, The anonymous Functions		10
	Total		28
Reference Text/Materials: 1. Mark Lutz, Learning Python, 5th Edition, O'Reilly Media Publishers 2. Think Python: How to think like a Computer Scientist by Allen Downey, 2nd edition, O'Reilly Publications: https://greenteapress.com/wp/think-python-2e/ https://www.w3schools.com/python			

PAPER CODE: UCSH-203	Software Lab in Python-II	CREDITS: 1((0-0-2)
Course Category:	SEC	Total Periods: 28
Prerequisite (If any): Software Lab in Python -I		
The course aims to: introduce students to advanced features of Python, such as complex data types, classes, and file handling, introduction to object-oriented Programming and data visualization		
Course Outcomes: At the end of the course, the student will be able to: • write and execute simple programs using strings, dictionaries, and tuples, and on classes and file handling in Python • create classes and objects • sketch graphs using appropriate Python libraries, and interpret them for a better understanding • develop proficiency in creating and manipulating NumPy arrays. • use mathematical and statistical operations on arrays.		
Unit	Description	Periods
I	Modules and Classes: How to write a module in Python, writing various customized modules for the purpose of reuse. Also introducing the concept of object-oriented programming using classes, inheritance, and encapsulation. Understanding List, Dictionary and Tuple Operations	10
II	Introduction to matplotlib (Using the pyplot submodule) figure, axes, subplot plot(), scatter(), show(), labels, legends, titles, styles, ticks, dynamically updating curves, saving graphs	6
III	Introduction to numpy The numpy array – properties: size, shape, ndim, dtype, creating arrays, zeros, one(), full(), fill(), arange(), linspace(), logspace(), identity(), eye(), astype(), indexing and slicing arrays (view versus copy), important array methods- reshape(), ravel(), flatten(), hstack() and vstack(), Aggregate functions- p.sum(), np.prod(), np.mean(), np.std(), np.var(), np.min(), np.max(), np.argmax(), np.argmin(), np.argmax()	12
	Total	28
Reference Text/Materials: Reference Text/Materials: 1. Think Python: How to think like a Computer Scientist by Allen Downey, 2nd edition, O'Reilly Publications: https://greenteapress.com/wp/think-python-2e/ 2. Mark Lutz, Learning Python, 5th Edition, O'Reilly Media Publishers 3. https://www.w3resource.com/python/python-tutorial.php		

Course Title	Software Lab in Data Visualization	L	T	P	C	Periods
Course Code:	UCSH-303	0	0	4	2	56
Course Category	SEC					
Course Objectives: To make Data Storytelling through visualization of the data						
Course Outcomes: Develop the skill set to 1. Distinguish qualitative and quantitative data 2. Make inferences out of the different plots 3. Develop story out of the data						
Prerequisites for the Course: Basic Python/R						
Unit	Title Unit Contents					Periods
1	Numbers that Summarize the data	Measures of Average, Measures of Variance, Measures of correlation and Measures of Ratio.				14
2	Fundamental Variations of Graphs	A Brief History of Graphs; Graphical means for Qualitative data; Visual attributes: Lines, Histograms, Bar plot, Scatter plot and Box-Plots and inferences through examples				14
3	General Design Principles for Communication	Organizing, Highlighting, Integration, Table Design, General Graph Design, Multi-Variable display;				14
4	Case Studies	Take different data sets and make data storytelling with visualization.				14
		Total				56
*For implementation: Instructor can use any visualization framework Matplotlib/Seaborn/Plotly, ggplot2, Tableau, PowerBI, or any other						
Key Text: 1. Stephen Few, Show me the numbers: Designing tables and graphs to enlighten, Second Edition, Analytic Press, 2012 Chapters - 1 to 11 References: 2. Cole Nussbaumer Knafllic, Storytelling with Data, John Wiley & Sons, Inc. - 2015 3. Matplotlib Documentation, Retrieved June 30, 2022 from https://matplotlib.org/3.5.1/plot_CourseCategories/index.html 4. Seaborn Documentation, Retrieved June 30, 2022 from https://seaborn.pydata.org/introduction.ht						

Course Title	Software Lab in SQL		L	T	P	C	P
Course Code	UCSH-404		0	0	4	2	56
Course Category	SEC						
Course Objectives: It introduces basic query writing using SQL and MYSQL workbench. Course Outcome: Develop skill set to 1. Write simple DDL statements using MySQL and MySQL Workbench 2. Write simple DML statements MySQL and MySQL Workbench 3. Implement triggers, logs along with locks							
Prerequisites for the Course: NA							
Unit	Topic	Contents				Periods	
1	SQL DDL (data Definition language)	Working on MYSQL command prompt as well as MYSQL workbench : To Create, insert ,delete, drop, alter, set, rename, operations on the databases (at least 6 tables from the databases mentioned) using with keys viz., primary, foreign and basic data Course Categories viz., varchar, int, float, date, time etc.				20	
2	SQL DML (Data manipulation languages)	MYSQL command prompt as well as MYSQL workbench : Using Select, from, where for single as well as multiples tables, Using Aliases, distinct,*,Union, minus and intersect erations, Nested queries, Exists function, use of NULL, Using aggregate functions viz, max, min and avg.				20	
3	Triggers	Triggers, Locks and Logs				16	
Total						56	
References: 1. MYSQL reference manual 5.7. https://downloads.mysql.com/docs/workbench-en.pdf							

Multi-Disciplinary Courses

PAPER CODE: UMAT-001	Fundamentals of Statistics with Excel	CREDITS: 3 (2-0-1)
Course Category:	MDC	Total Periods: 42
Pre-requisite (If any): 10+2 any Discipline		
Course Objectives: The objective is to give basic knowledge of the subject and familiarity with techniques to use them effectively. Knowledge of Mathematics at the Senior Secondary level will be an advantage but not necessary for offering this course. The scope of Statistics has been well recognized over the years but has very much expanded in recent years and with the impressive progress in		
Course Outcomes: At the end of the course, the student will be able to: 1. Recognize and differentiate between key terms. 2. Apply various types of sampling methods to data collection. 3. Create and interpret frequency tables. 4. Display data graphically and interpret graphs: stemplots, histograms, and box plots. 5. Recognize, describe, and calculate the measures of location of data: quartiles and percentiles. 6. Recognize, describe, and calculate the measures of the center of data: mean, median, and mode. 7. Recognize, describe, and calculate the measures of the spread of data: variance, standard deviation, and range.		
Unit	Description	Periods
1	Statistics: What is statistics, Uses and misuses of statistics, Introduction to basic terms, Measurability and variability, Relationship between probability and statistics?	4
2	Descriptive Analysis and Presentation of Single Variable Data: Data collection, Measures of central tendency, Measures of Dispersion, Graphic Presentation and Measures of Position.	14
3	Descriptive Analysis and Presentation of Single Variable Data (Continued)	12

	Frequency distributions, Graphic presentations, Calculations using frequency distribution, Interpreting and understanding Standard Deviation, The art of Statistical Deception.	
4	Descriptive Analysis and Presentation of Bivariate Data Bivariate Data, Linear Correlation, Linear Regression	12
Key Text: 1. Robert R. Johnson, Elementary Statistics, Duxbury Press, North Scituate, Massachusetts, 1976, 2 nd edition Chapters: 1,2,3, 4		
Suggested Readings: 1. Mario F. Triola, Elementary Statistics, Addison-Wesley, 7 th Edition, 1997. 2. Robert N. Goldman, Joel S. Weinberg, Statistics An Introduction, Prentice-Hall Publication 1985.		

PAPER CODE: UMAT-002	The Language of Mathematics	CREDITS: 3 (3-0-0)
Course Category:	MDC	Total Periods: 42
Prerequisite (If any): 10+2 Mathematics		
Course Objectives: The course aims to: 1. Give a basic understanding of working with numbers which include Modular Arithmetic, the Principle of Induction, and the usage of prime numbers. 2. Give a basic understanding of Geometry, Euclid's elements, conic sections, non-Euclidean geometry, and higher dimensions 3. Give a basic understanding of Probability, experiments, and chance		
Course Outcomes: At the end of the course, the student will be able to: 1. Start any abstract course on Number Theory 2. Read courses on Applied and Theoretical Probability.		
Unit	Description	Periods
I	Why Do Numbers Count? You can count on them;These days, children do it before they are five;A token advance ; Symbolic Progress; For a long time, it was all Greek;A fatal law is discovered; Here's looking at Euclid; Numbers in Prime condition; Prime order;The child Genius; Gauss's clock Arithmetic;The Great Amateur;Taking the prime test;Keeping Secrets; Easy to Guess, Hard to prove;Fermat's Last Theorem;The Fermat Saga Begins;The Domino Effect	18

II	Mathematics Gets into Shape Everyone is a geometer;Measuring the Earth;Things that Euclid missed;Euclid in his elements;Geometry as a theory of Everything;Slicing through Cones;A fly on the ceiling Squaring the circle and other impossibilities;The startling discovery of non-Euclidean geometries;The geometry of Renaissance artists;Beyond the third dimension	18	
III	How do mathematicians figure the odds? Who gets Heaven?;Figuring the odds;The geometric patterns of chance;In Mathematics we trust;The Amazing Bernoulli;Fear of Flying;Ringing in the bell curve;The Reverend Bayes;Enter the Average Man;The abstract Patterns of chance;Using Mathematics to choose your options	20	
Key Text:		Topics covered	Prescribed chapter/ Sections
1. Keith Devlin, The Language of Mathematics (Making the invisible visible), W H Freeman and Company, 2000, 1 st edition		Numbers, Geometry and Probability	Chapter: 1, 4, 7
Suggested Readings: 1. Richard Courant and Herbert Robbins, what is Mathematics? Oxford University Press, 1941, 2 nd Edition			

PAPER CODE: UMAT-003	FUNDAMENTALS OF ACTUARIAL MATHEMATICS	CREDITS: 3 (3-0-0)
Course Category:	MDC	Total Periods: 42
Pre-requisite (If any): NA		
Course Objectives: The aim of the Actuarial Mathematics subject is to provide a grounding in financial mathematics and its simple applications		
Course Outcomes: At the end of the course, the student will be able to: 1. Understand and be able to perform calculations relating to present value, current value, and accumulated value. 2. Calculate present value, current value, and accumulated value for sequences of non-contingent payments. 3. Understand key concepts concerning loans and how to perform related calculations. 4. Understand key concepts concerning bonds, and how to perform related calculations 5. Understand key concepts concerning yield curves, rates of return, measures of duration and convexity, cash flow matching and immunization, and how to perform related calculations.		

Unit	Description	Periods
I	Time Value of Money a. Define and recognize the definitions of the following terms: interest rate (rate of interest), simple interest, compound interest, accumulation function, future value, current value, present value, net present value, discount factor, discount rate (rate of discount), convertible monthly, nominal rate, effective rate, inflation and real rate of interest, force of interest, equation of value. b. Given any three of interest rate, period of time, present value, and future value, calculate the remaining item using simple or compound interest. Solve time value of money equations involving variable force of interest. c. Given any one of the effective interest rate, the nominal interest rate convertible m-thly, the effective discount rate, the nominal discount rate convertible m-thly, or the force of interest, calculate any of the other items. d. Write the equation of value given a set of cash flows and an interest rate.	7
II	Annuities/cash flows with non-contingent payments a. Define and recognize the definitions of the following terms: annuity-immediate, annuity due, perpetuity, payable m-thly or payable continuously, level payment annuity, arithmetic increasing/decreasing annuity, geometric increasing/decreasing annuity, term of annuity. b. For each of the following types of annuity/cash flows, given sufficient information of immediate or due, present value, future value, current value, interest rate, payment amount, and term of annuity, calculate any remaining item. - o Level annuity, finite term. - o Level perpetuity. - o Non-level annuities/cash flows. Arithmetic progression, finite term and perpetuity. Geometric progression, finite term and perpetuity. Other non-level annuities/cash flows.	13
III	Loans a. Define and recognize the definitions of the following terms: principal, interest, term of loan, outstanding balance, final payment (drop payment, balloon payment), amortization. b. Calculate: • The missing item, given any four of: term of loan, interest rate, payment amount, payment period, principal. • The outstanding balance at any point in time. • The amount of interest and principal repayment in a given payment. • Similar calculations to the above when refinancing is involved	11
IV	Bonds a. Define and recognize the definitions of the following terms: price, book value, market value, amortization of premium, accumulation of discount, redemption value, par value/face value, yield rate, coupon, coupon rate, term of bond, callable/non-callable, call price, call premium, accumulated value with reinvestment of coupons.	11

	b. Given sufficient partial information about the items listed below, calculate any of the remaining items • Price, book value, market value, accumulated value with reinvestment of coupons, amortization of premium, accumulation of discount. (Note that valuation of bonds between coupon payment dates will not be covered). • Redemption value, face value, Yield rate, Coupon, coupon rate. • Term of bond, point in time that a bond has a given book value, amortization of premium, or accumulation of discount. Calculate the price of a callable bond to achieve a specified minimum yield.	
Key Text: Broverman, S.A., Mathematics of Investment and Credit (Seventh Edition), 2017, ACTEX Publications, ISBN 978-1-63588-221-6		
Prescribed Chapters/ Sections : Chapter 1 (excluding 1.2.1 and 1.8), Chapter 2 (excluding 2.3.1.2, 2.4.2, 2.4.3 and 2.4.5), Chapter 3 (excluding 3.2.1, 3.2.2, 3.3, and 3.4), Chapter 4 (excluding 4.1.3), Chapter 5 (excluding 5.2, the investment year method portion of 5.3.1, and excluding all of 5.3.2, 5.3.3 and 5.3.4), Chapter 6 (excluding 6.2 and 6.4), Chapter 7 (excluding 7.1.3, 7.1.6 and 7.3)		
Suggested Readings: 1. Bowers, N. L.; Gerber, H. U.; Hickman, J. C. et al., In Actuarial Mathematics, 2nd ed. Society of Actuaries, 1997. 753 pages. ISBN: 9780938959465. 2. Butcher, M. V., Mathematics of compound interest, Nesbitt, C. J. Ulrich's Books, 1971. 324 pages. ISBN: 9780960300013. 3. Ingersoll, J. E. Rowman & Littlefield, 1987, Theory of financial decision making. 474 pages. ISBN: 9780847673599. 4. Kellison, S. G., The theory of interest. 3rd ed. Irwin, 2008. 463 pages. ISBN: 9780073382449. 5. Gerber, H. U., Life insurance mathematics, 3rd ed. Springer; Swiss Association of Actuaries, 1997. 217 pages. ISBN: 9783540622420.		